Investigating the Impact of COVID-19 Restrictions on Air Pollution Levels in Massachusetts Cities

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1 Introduction

Residents of urban areas live among diverse primary sources of air pollution, including power plants, waste incinerators, and high-density transportation networks. The concentration of vehicle and air traffic in U.S. cities exposes residents to elevated levels of Particulate Matter ($PM_{2.5}$) [1]. These fine particles, often byproducts of combustion, are small enough to remain suspended for long periods and penetrate deep into the respiratory and circulatory systems upon inhalation, causing significant respiratory, cardiovascular, and neurological harm [2, 3].

The onset of the COVID-19 pandemic in March 2020 created a unique opportunity to observe air quality shifts under extreme conditions. As lockdowns and stay-at-home advisories were implemented, human activity plummeted. Globally, NO_2 levels dropped by approximately $20 - 30\%$, while U.S. cities saw substantial reductions in $PM_{2.5}$ emissions as air passenger volume dropped by 60.2% and vehicular traffic reached historic lows[4, 5].

While transportation emissions are a well-documented driver of climate change and public health inequalities, most pandemic-era research has focused exclusively on major metropolitan hubs. There remains a critical gap in understanding how these disruptions affected air quality in smaller towns and secondary cities. This study builds upon existing $PM_{2.5}$ research by evaluating the impact of Massachusetts' COVID-19 State of Emergency (SOE) across four cities with varied population sizes, highway densities, and proximity to major airports. By comparing these diverse urban environments, we aim to provide evidence for environmental policies that target specific local emission sources more effectively.

2 Background

Urban air pollution from transportation, including both roadway traffic and aviation, is a major public health concern due to its contribution to particulate matter exposure and related respiratory and cardiovascular health effects. For example, reductions in vehicular traffic during major urban events such as the 1966 Atlanta Olympics and the 2008 Beijing Olympics were associated with substantial decreases in traffic-related emissions, including peak ozone reductions of 28% and traffic-related emissions decreases of 21-61%, respectively [6, 7].

The COVID-19 pandemic created an unprecedented natural experiment to quantify the contributions of both everyday transportation and long-distance travel. Transportation restrictions implemented during the state-of-emergency periods significantly reduced traffic volumes and flight activity across the globe. For instance, nearly 96% of aviation operations were disrupted in the U.S., providing a unique opportunity to study the impact of aviation and traffic reductions on air pollution levels [8]. Similarly, studies of near-roadway neighborhoods in Somerville, MA, demonstrated reductions in air pollutants during the lockdown period, directly linking traffic reductions to improvements in local air quality scales not captured by regional regulatory monitoring networks [9]. Although this study did not use measurements of $PM_{2.5}$, its findings on the reduction of other air pollutants due to travel restrictions suggest that decreased human mobility has the potential to impact traffic-related air quality in general.

A 2022 study on $PM_{2.5}$ levels during the COVID-19 lockdown utilized deep learning methods to find that daily $PM_{2.5}$ levels in several U.S. cities, including Boston, decreased coinciding with a decrease in human mobility. Areas with a larger decrease in human mobility saw the most significant improvements in air quality compared to those where individuals normally remain in their residential areas longer, indicating that air quality improvements as a result of the lockdown are more likely to have occurred in densely populated and socially active areas. In Boston, the amount of time people stayed in their residential areas increased by 20.43% in March to May 2020 compared to January to February 2020 [5].

These studies collectively build a better understanding for how different transportation sectors like aviation versus road traffic contribute to local air pollution in densely populated urban areas, and how these contributions vary under extreme changes in human activity. The COVID-19 pandemic offers a rare opportunity to address these gaps by providing a prolonged period of reductions in aviation and ground traffic, enabling more precise quantification of source-specific impacts on $PM_{2.5}$.

This study extends prior research by leveraging a long-term air monitoring dataset in four Massachusetts cities to evaluate the impact of state-imposed travel restrictions during the COVID-19 State of Emergency on $PM_{2.5}$ concentrations. The findings have practical implications for urban air quality management and public health, informing policies for emissions reductions in cities. Moreover, the study provides methodological contributions by demonstrating how natural experiments can quantify sector-specific contributions to particulate matter concentrations.

3 Methods

3.1 Data Collection

The primary dataset in this analysis was obtained from the United States Environmental Protection Agency Air Quality System (AQS), which provides publicly available measurements of ambient air pollutants across the United States. The EPA collects these data to monitor compliance with national air quality standards and to support environmental and public health research.

The dataset consists of daily average measurements of particulate matter with aerodynamic diameter less than 2.5 micrometers ($PM_{2.5}$), recorded at fixed-site monitoring stations across multiple cities in Massachusetts. These monitors capture local ambient pollution levels influenced by urban activity, regional emissions, and environmental conditions. Cities included in this study represent a range of urban and suburban environments, allowing for comparison of pollution patterns across areas of varying traffic.

Each observation includes the date of measurement and the daily mean $PM_{2.5}$ concentration, which serves as the primary variable of interest. This variable is interpreted as an indicator of overall air pollution exposure on a given day. Monitoring sites are identified using station metadata, including location and site name.

$PM_{2.5}$ measurements were collected on a systematic 1-in-3 day sampling schedule, resulting in observations approximately every three days. This sampling design is commonly used in regulatory monitoring by the EPA to balance temporal resolution with operational constraints. This frequency is sufficient to capture broader temporal trends and seasonal patterns, although very short-term fluctuations may not be fully represented.

The dataset spans multiple years, including the period before, during, and after the COVID-19 State of Emergency beginning in March 2020. This time frame allows for direct comparison of seasonal $PM_{2.5}$ patterns under typical conditions and during a period of significant disruption to human activity.

3.2 Variable Creation

Data from four Massachusetts cities (Boston, Worcester, Brockton, and North Adams) were obtained from the EPA's Air Quality System for the years 2018, 2019, 2020, 2021, and 2022. Prior to analysis, the dataset was filtered to ensure consistency across monitoring sites and time. Each city had multiple monitoring sites. For each city and year, we selected the parameter occurrence code (POC) of the monitor that recorded a full year of $PM_{2.5}$ data to ensure consistency in temporal coverage. Specifically, Boston used POC 1 for 2018 to 2021, and POC 3 for 2022, while all other cities used POC 3 across all years.

From the raw datasets, we selected only the date of measurement and the daily mean $PM_{2.5}$ concentration. The separate yearly datasets were merged by date for all cities to produce a single time series dataset for analysis.

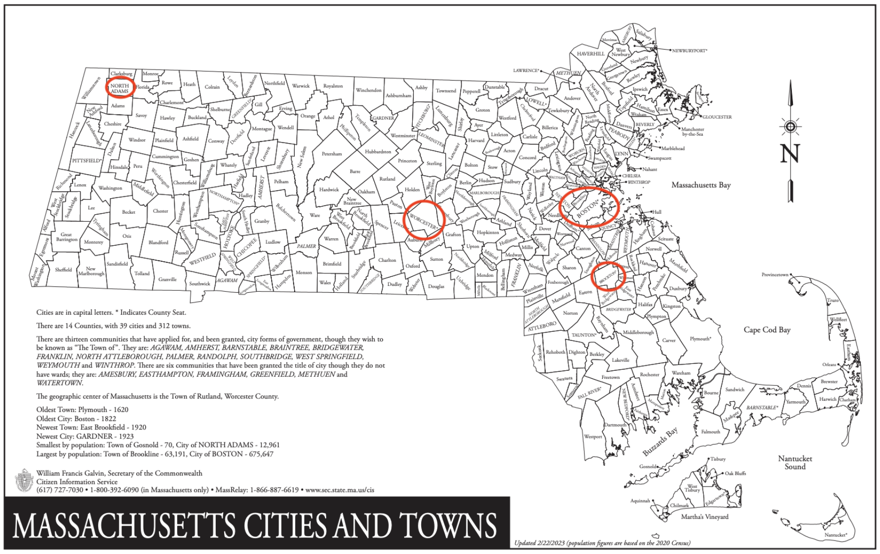

Figure 1: Map of Boston, Brockton, North Adams, and Worcester, Circled in Red

We used year-month-day date formats to facilitate plotting and trend analyses. Missing dates were retained as NA to maintain alignment across cities, particularly for monitors that recorded measurements every three days, such as Boston in 2019. Finally, for visualization and interactive presentation, the data was merged to include the daily $PM_{2.5}$ concentration for every three days from 2018 to 2022 for each of the four cities. This format allows for straightforward comparison of seasonal trends across cities and years.

4 Results

The primary objective of this study is to examine how seasonal trends in $PM_{2.5}$ concentrations were affected during the COVID-19 State of Emergency. Rather than explicitly modeling individual sources of pollution such as traffic or aviation, this analysis focuses on identifying shifts in overall seasonal patterns across four Massachusetts cities.

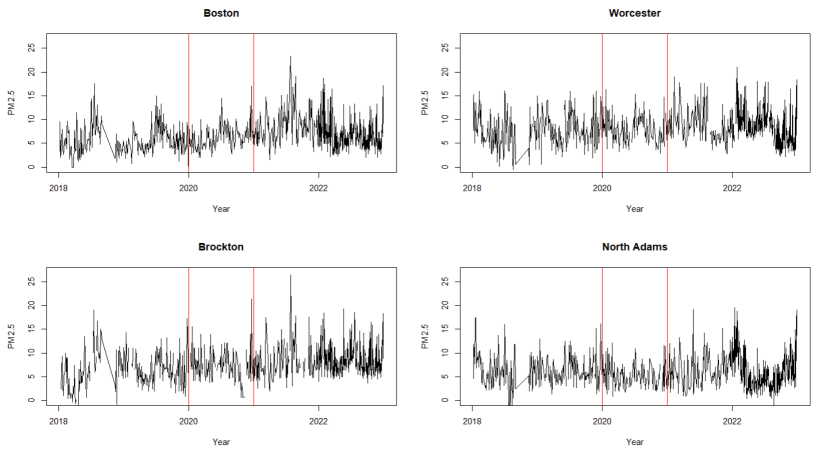

Figure 2: Plots of the Daily Mean $PM_{2.5}$ Concentration for Each City

We began with exploratory time series visualization to identify seasonal trends, long-term patterns, and abrupt changes in $PM_{2.5}$ concentrations. Line plots were used to compare patterns across years, with particular attention given to deviations occurring during the 2020 restriction period. First, we plotted the daily mean $PM_{2.5}$ concentration for each city, with red bars representing the year of 2020 (Figure 2). Then, seasonality was assessed by examining recurring patterns within each year, comparing the amplitude and timing of peaks between pre-pandemic and pandemic periods to identify potential disruptions.We plotted the daily mean $PM_{2.5}$ concentration for each city for each year to see the relationship over the year, shown in Figure 2. It is important to note that there is missing data for Figure 2, September and October of 2018 for all four cities, so the series for 2018 connects the last non-NA value date in August to the first non-NA value in October of 2018. Figure 3 illustrates a more granular view of the daily PM2.5 concentrations by overlaying years 2018 through 2022, with the black line representing the year 2020.

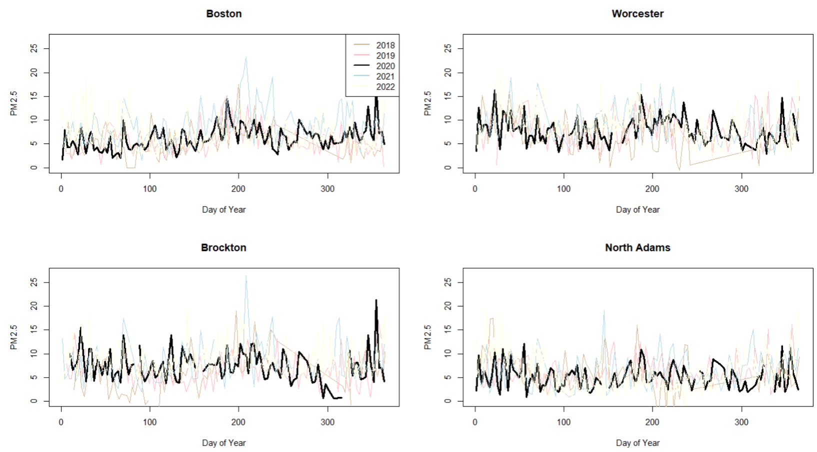

Figure 3: Daily Mean $PM_{2.5}$ Concentration for Each City by Year

To capture non-linear trajectory of air quality during the pandemic by city, we employed regression splines as seen in Figure 4. Specifically, quadratic splines were utilized with knots placed to isolate the 2020 window. This allows the model to "bend" and account for the lockdown phase and subsequent period of pollution levels. The models reveal a clear divide in how different urban environments responded to the events of 2020. The summary tables for the regressions spline fits can be seen in Table 1.

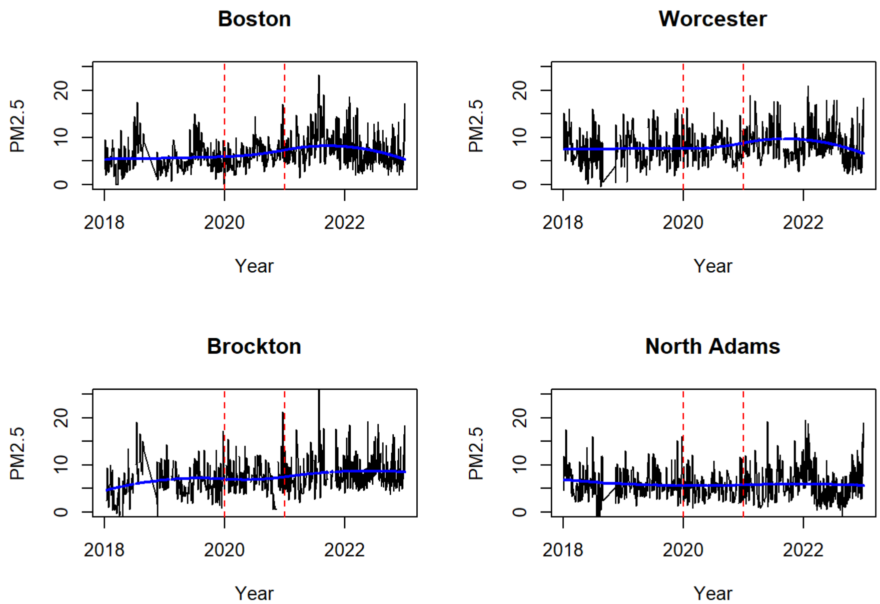

Figure 4: Regression Splines for Each City

The model for Boston provided the strongest evidence of a systematic change in pollution trajectory, explaining the highest proportion of variance ($R^2 = 0.14$, $p < 0.001$). While the linear and quadratic terms were not individually significant, the spline knots revealed critical transition points. The first knot was significant, but the second knot, representing late 2020, was highly significant ($p < 0.001$). This indicates a rigorous downward bend in $PM_{2.5}$ concentrations following the initial shock of the state of emergency period.

Brockton followed a similar, slightly less pronounced pattern ($R^2 = 0.075$, $p < 0.001$). Unlike Boston, Brockton showed significant initial growth in the linear term (p =0.004), but both the first and second knots were statistically significant, confirming that the spline model successfully captured a unique 2020 trajectory in this area.

In contrast, the air quality trajectories in Worcester were far less defined. The model fit was poor ($R^2 = 0.032$), and none of the spline-specific terms reached the standard threshold for significance. The first and second knots failed to confirm a detectable shift in the $PM_{2.5}$ trend. North Adams showed the weakest overall fit ($R^2 = 0.012$), with the model failing to reach the standard 0.05 significance level ($p = 0.1639$). These results imply that in Worcester, pollution levels were dominated by random fluctuations and short-lived spikes rather than a broad, human induced curve.

Across all four models, the residuals exhibited large positive maxima, particularly in Brockton (18.22) and Boston (14.69). These values correspond to the

Table 1: Regression Spline Results Across Cities

	Boston	**Worcester**	**Brockton**	**North Adams**
Intercept	5.001 (0.579)	7.761 (0.638)	4.495 (0.669)	6.987 (0.616)
t	0.010 (0.010)	-0.005 (0.011)	0.036 (0.013)	-0.011 (0.011)
t^2	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
Knot 1	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Knot 2	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Model Fit Statistics				
Observations	550	528	511	535
R^2	0.130	0.033	0.075	0.012
Adj. R^2	0.124	0.025	0.068	0.005
AIC	2811.6	2785.8	2720.4	2789.4
BIC	2837.5	2811.5	2745.8	2815.1
Log Likelihood	-1399.822	-1386.918	-1354.193	-1388.699
F Statistic	20.360	4.400	10.248	1.635
RMSE	3.08	3.35	3.43	3.24

Note: Entries report coefficient estimates with standard errors in parentheses.

most extreme deviations between observed and fitted pollution levels, indicating occasional high-magnitude events not fully captured by the spline model.

The low overall R^2 values across the board, ranging from 0.012 in North Adams to 0.13 in Boston, are consistent with environmental time-series data, which are heavily influenced by stochastic weather patterns and localized events. However, the high F-statistics and significant p-values for the Boston and Brockton models confirm that, despite the daily volatility, there was a statistically valid underlying trend unique to those cities during the 2020 period.

To evaluate the accuracy of the spline fit, we examined the Autocorrelation Function (ACF) and the residuals over time, and performed the Ljung-Box test. These tests assess whether the model successfully reduced the data to white noise or if significant temporal dependencies remain. The ACF plots for all four cities demonstrate significant positive autocorrelation at early lags, confirming that $PM_{2.5}$ concentrations are not independent day-to-day (Figure 5). Boston exhibits the highest degree of persistence, with significant autocorrelation extending beyond 15 days, whereas North Adams shows a more rapid decay, indicating that pollution events there are shorter-lived. This suggests that pollution in Boston is persistent, possibly due to its dense urban environment or consistent traffic patterns that take a long time to dissipate from the system. The North Adams and Worcester ACF plots indicate that pollution events in these areas are more transient. They happen and then dissipate relatively quickly compared to Boston. Also, the presence of subtle peaks at seven-day intervals in the Boston and Brockton plots suggests a latent weekly seasonality, most likely due to anthropogenic cycles, like commuter traffic.

Looking at the residuals over time, you can see exactly where our spline

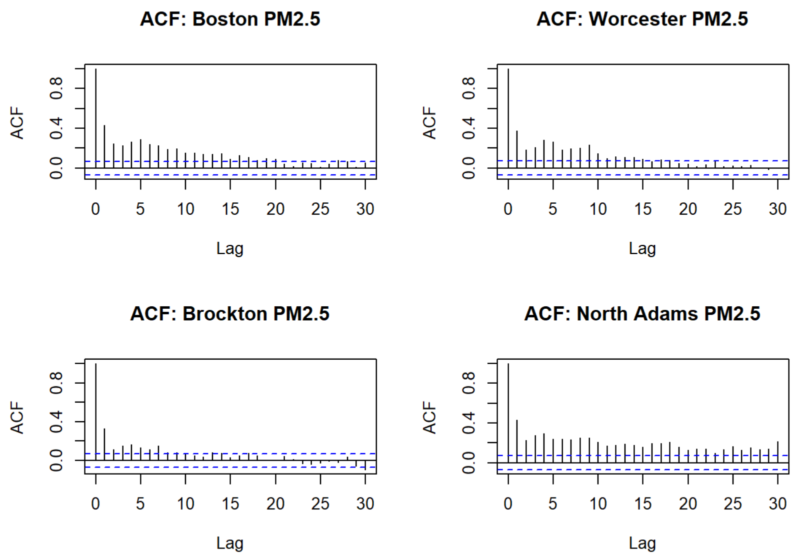

Figure 5: ACF Plot for Each Massachusetts City

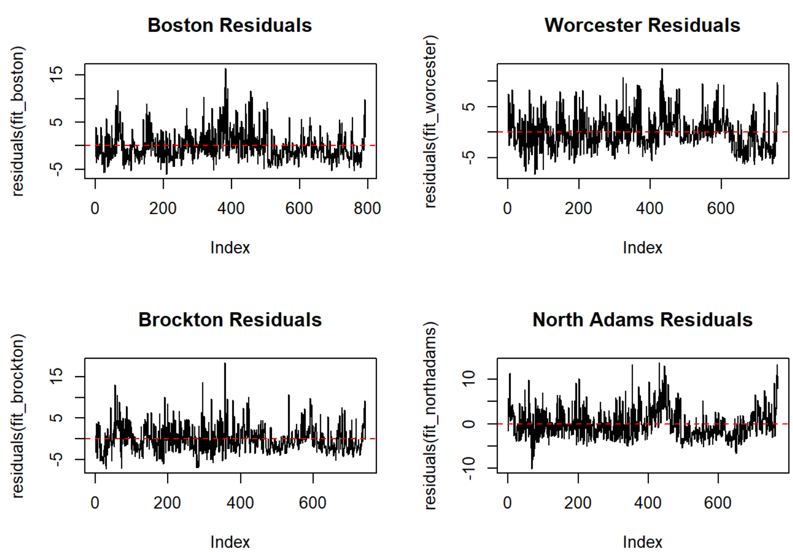

Figure 6: Residuals Over Time for Each Massachusetts City

model succeeded and where it struggled to keep up with the actual air quality data. The residual plots for all four cities exhibit distinct patterns of non-randomness, primarily characterized by temporal clustering and positive skewness. While the regression splines effectively modeled the long-term trend, the residuals show significant leftover structure. In all plots, there are several periods where the residuals stay entirely above or below zero for 10 to 20 days at a time. This is a visual confirmation of the autocorrelation we found earlier. It means our spline is missing short-term trends. In Brockton, there is a spike that hits nearly 20 units above the prediction. These represent specific daily events where the air quality was significantly worse than the average trend for that time of year. Since our spline model is built to be a smooth curve, it ignores these spikes, leaving them behind as large residuals. Furthermore, the high-magnitude positive spikes observed across all cities suggest that the $PM_{2.5}$ data is influenced by short-term atmospheric events that the smooth spline basis functions are not mathematically equipped to capture.

Table 2: Ljung–Box Test for Residual Autocorrelation

City	χ^2 Statistic	df	p-value
Boston	186.19	16	$< 2.2 \times 10^{-16}$
Worcester	84.003	16	3.135×10^{-11}
Brockton	74.384	16	1.682×10^{-9}
North Adams	116.33	16	$< 2.2 \times 10^{-16}$

The Ljung-Box test results in Table 2 across all four study sites ($p < 0.001$) provide strong statistical evidence that the residuals from the regression spline models are not independent. The high chi-squared values, particularly in the Boston model ($\chi^2 = 239.38$), indicate that substantial autocorrelation remains unaddressed. This suggests that while the splines effectively mapped the non-linear impact of the 2020 period on $PM_{2.5}$ levels, a standard ordinary least squares (OLS) approach lacks the temporal complexity to fully resolve the daily stochastic nature of air pollution.

To address the limitations identified in the initial spline models, specifically high autocorrelation and non-independence of residuals, we performed additional diagnostic testing and employed robust estimation techniques.

First, the Augmented Dickey-Fuller (ADF) test was applied to the $PM_{2.5}$ time series for all four cities. Each city yielded a p-value of less than 0.01, allowing us to reject the null hypothesis of a unit root. This confirms that the data is stationary, providing a valid foundation for time series regression and ensuring that the observed temporal variations are not the result of a random walk.

To account for the identified autocorrelation and potential heteroskedasticity in the pollution data, we employed Heteroskedasticity and Autocorrelation Robust (HAR) estimators. Rather than altering the spline coefficients themselves, the HAR approach provides a heteroskedasticity and autocorrelation consistent covariance matrix. This approach ensures that the standard errors and resulting p-values remain valid even when standard OLS assumptions regarding the error terms are violated.

Table 3: Robust Regression Results Across Cities

	Boston	**Brockton**	**North Adams**	**Worcester**
Estimates (Std. Errors)				
Intercept	5.00 (56.92)	4.49 (40.61)	6.99 (66.56)	7.76 (47.04)
t	0.01 (0.84)	0.04 (0.63)	-0.01 (1.14)	-0.00 (0.76)
t^2	-0.00 (0.00)	-0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Knot 1	0.00 (0.00)	0.00 (0.00)	0.00 (0.01)	0.00 (0.00)
Knot 2	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)	-0.00 (0.00)
t-values (p-values)				
Intercept	0.09 ($\geq$0.10)	0.11 ($\geq$0.10)	0.10 ($\geq$0.10)	0.17 ($\geq$0.10)
t	0.01 ($\geq$0.10)	0.06 ($\geq$0.10)	-0.01 ($\geq$0.10)	-0.01 ($\geq$0.10)
t^2	-0.01 ($\geq$0.10)	-0.05 ($\geq$0.10)	0.01 ($\geq$0.10)	0.01 ($\geq$0.10)
Knot 1	0.04 ($\geq$0.10)	0.07 ($\geq$0.10)	0.00 ($\geq$0.10)	0.00 ($\geq$0.10)
Knot 2	-0.15 ($\geq$0.10)	-0.11 ($\geq$0.10)	-0.03 ($\geq$0.10)	-0.07 ($\geq$0.10)

Table 3 reports the robust regression estimates, associated standard errors, t-statistics, and p-values for each city. While the point estimates remain similar to the original spline models, the standard errors are substantially larger, resulting in uniformly small t-statistics and p-values exceeding 0.10 across all coefficients. This indicates that individual spline components are not statistically significant once temporal dependence is accounted for.

While the individual t-tests for specific spline knots often yielded p-values greater than 0.10 due to increased robust standard errors, the Wald F-test was employed to assess the collective significance of all temporal terms. The robust analysis in Table 4 reveals a clear separation in model performances across the study sites.

In Boston and Brockton, despite the inflation of standard errors for individual predictors, the robust F-test confirmed that the temporal variables are collectively significant. This indicates that the non-linear trends captured by the splines are statistically rigorous and not merely a result of auto correlated noise. In these high-density areas, the signal of air quality change remains dominant even after conservative adjustments.

Table 4: Robust F-Test

	F Statistic	P-Value
Boston	23.13	<0.01*
Brockton	20.62	<0.01*
North Adams	1.70	>=0.10
Worcester	9.29	<0.10

The model for Worcester exhibits marginal significance with a p-value less than 0.10. While the splines visually suggest a lockdown-related bend, the significance is weakened when accounting for daily stochastic fluctuations, suggesting a trend that is present but less systematic than those in the larger urban centers.

The model for North Adams yielded a p-value greater than 0.10, failing to reject the null hypothesis. This confirms that the temporal fluctuations in North Adams are indistinguishable from noise once the high variability of the data is properly modeled.

These findings clarify where the signal of air quality change is strongest. While the splines initially showed visual bends for all cities, the robust testing suggests that only Boston and Brockton show consistent, statistically significant temporal trends that survive adjustment for autocorrelation. This suggests that air quality in these areas is driven by steady urban patterns, like daily traffic, rather than random, short-lived spikes seen in North Adams.

5 Discussion

The primary goal of this analysis was to reveal how trends in daily $PM_{2.5}$ concentration shifted across four Massachusetts cities during the 2020 period of reduced urban activity. By analyzing five years of EPA air quality data through quadratic regression spline models, we aimed to identify distinct atmospheric trends before, during, and after the pandemic's onset.

Our findings reveal a significant divide between dense urban centers and smaller cities. Boston and Brockton both exhibited statistically rigorous temporal trends. Even after applying conservative robust adjustments for autocorrelations and heteroskedasticity, the joint significance of the temporal variables in these cities remained high. This suggests that in high-volume areas, the shift in pollution was a systematic departure from previous years rather than a random fluctuation.

In contrast, the results for Worcester and North Adams suggest that air quality in these areas is driven more by baseline emissions or stochastic events. While Worcester's model showed marginal significance ($p < 0.10$), North Adams lacked any statistically significant trend ($p > 0.10$). In these regions, pollution levels appear more susceptible to short-lived spikes, potentially from residential heating or localized industrial activity, that were not statistically altered by the pandemic. This implies that while major urban hubs see immediate air quality benefits from reduced travel, small or more rural cities may require different environmental strategies to address their primary pollution sources.

We can conclude that Boston's air quality most clearly displays a consistently unique trend in 2020 compared to the surrounding years of 2018-2019 and 2021-2022. Brockton followed a similar trajectory, with its overall model performance confirming shifts that align with the timeline of urban restrictions. Conversely, air quality trends in Worcester and North Adams did not show the same level of resilience to robust statistical testing. This suggests that the root of emissions in these cities may come from processes that are less reliant on social urban activity and commuter travel.

The decrease in urban activity that occurred due to COVID-19 restrictions in 2020 likely led to measurable air quality improvements in high-traffic areas like Boston and Brockton. This demonstrates the role that regulating air pollution emissions from industry and traffic can play in reducing the city's $PM_{2.5}$ concentrations. This information could be extremely useful for policymakers who can set stricter regulations on sources of air pollution, and work towards a healthier air quality for the members of their community.

5.1 Limitations

While the analysis successfully isolated temporal trends in high-density areas, several limitations inherent to atmospheric data and the study's scope must be acknowledged.

The regression spline models exhibited high residuals and significant autocorrelation. This indicates that while a smooth curve can capture the broad trend, it is mathematically insufficient to account for the high daily volatility of $PM_{2.5}$. These models may struggle to differentiate between a long-term reduction in human activity and extreme, short-lived spikes caused by localized events or specific industrial emissions.

Air quality can vary significantly at small spatial scales within cities due to proximity to roadways, industrial sites, and residential combustion sources. Because each city is represented by a limited number of EPA monitoring stations, the results may not fully capture within-city heterogeneity in pollution exposure. Additionally, the models do not incorporate meteorological variables such as temperature, wind speed, humidity, or atmospheric pressure, all of which strongly influence $PM_{2.5}$ dispersion and accumulation. The omission of these covariates limits the ability to separate anthropogenic effects from weather-driven variability.

Data were sourced from fixed-site monitors, which provide a single aggregate value for an entire city. Air quality can vary significantly at micro-spatial scales. Therefore, the specific location of a sensor may skew the data, failing to represent the true exposure levels of the city's broader population.

The EPA's 1-in-3 day sampling schedule effectively thins the time series. This lower temporal resolution may smooth over significant but brief population events that occurred during the lockdown, potentially underestimating the magnitude the State of Emergency impact.

5.2 Future Directions

To deepen the understanding of how policy-driven changes in human activity affect air quality in Massachusetts cities, future research should transition from descriptive trend analysis to causal and predictive modeling. To move beyond the idea of the "State of Emergency", future models should incorporate daily traffic counts and flight frequency as direct predictors. This would allow for a more precise quantification of how much each specific transportation sector contributes to $PM_{2.5}$ levels in different urban contexts. These models could help us understand why air quality may have changed or stayed relatively the same in different areas during 2020 when many restrictions limited urban activity. Air travel data, similarly, may be informative if considered as a predictor in modeling the change in air pollutant trends. Since these forms of travel were limited during 2020, it would provide insight into how air pollution may have changed on days when air travel and car traffic were much lower than average.

Future iterations of this work could employ ARIMA models to better handle the autocorrelation found in this study. This is a specialized time series method designed to handle the "memory" of weather and pollution patterns better than standard curves.

Instead of us telling the model when the lockdown started, we could use on-line change-point methods that allow the computer to look at the data and mathematically pick out the exact date the air quality trend shifted significantly. This would provide an unbiased look at when the State of Emergency actually began affecting the atmosphere.

Looking at chemicals like Nitrogen Dioxide or Black Carbon, which are more directly tied to tailpipe emissions than $PM_{2.5}$, might show an even clearer lockdown signal that isn't as influenced by background dust or wood-burning stoves.

References

[1] K. Riley et al. "A Systematic Review of The Impact of Commercial Aircraft Activity on Air Quality Near Airports". In: *City and Environment Interactions* 11 (2021). DOI: 10.1016/j.cacint.2021.100066. URL: https://doi.org/10.1016/j.cacint.2021.100066.

[2] State of Global Air. *PM2.5 Pollution Sources*. Accessed: 2026-05-05. 2025. URL: https://www.stateofglobalair.org/pollution-sources/pm25.

[3] X. Luo et al. "Impacts of urbanization process on PM2.5 pollution in "2+26" cities". In: *Journal of Cleaner Production* 284 (2021), p. 124761. DOI: 10.1016/j.jclepro.2020.124761. URL: https://doi.org/10.1016/j.jclepro.2020.124761.

[4] X. Sun, S. Wandelt, and A. Zhang. "COVID-19 Pandemic and Air Transportation: Summary of Recent Research, Policy Consideration and Future Research Directions". In: *Transportation Research Interdisciplinary Perspectives* 16 (2022), p. 100718. DOI: 10.1016/j.trip.2022.100718. URL: https://doi.org/10.1016/j.trip.2022.100718.

[5] M. Ghahremanloo et al. "A Comprehensive Study of the COVID-19 Impact on PM2.5 Levels over the Contiguous United States: A Deep Learning Approach". In: *Atmospheric Environment* 272 (2022), p. 118944. DOI: 10.1016/j.atmosenv.2022.118944. URL: https://doi.org/10.1016/j.atmosenv.2022.118944.

[6] Y. Wang et al. "Ozone air quality during the 2008 Beijing Olympics: effectiveness of emission restrictions". In: *Atmospheric Chemistry and Physics* 9 (2009), pp. 5237–5251. DOI: 10.5194/acp-9-5237-2009.

[7] M. S. Friedman et al. "Impact of Changes in Transportation and Commuting Behaviors During the 1996 Summer Olympic Games in Atlanta on Air Quality and Childhood Asthma". In: *JAMA* 285.7 (2001), pp. 897–905. DOI: 10.1001/jama.285.7.897.

[8] U.S. Government Accountability Office. *COVID-19 Pandemic: Observations on the Ongoing Recovery of the Aviation Industry*. GAO-22-104429. GAO, 2021. URL: https://www.gao.gov/products/gao-22-104429.

[9] N. Hudda et al. "Reductions in traffic-related black carbon and ultrafine particle number concentrations in an urban neighborhood during the COVID-19 pandemic". In: *The Science of the Total Environment* 742 (2020), p. 140931. DOI: 10.1016/j.scitotenv.2020.140931.

[10] Ryan Madsen. *Busiest Air Travel Year in 15 Years Puts Airport Investment in Focus*. Jan. 2026. URL: https://airportscouncil.org/2026/01/16/busiest-air-travel-year-in-15-years-puts-airport-investmentn-focus/.

[11] Airports Council International - North America. *Terminally Challenged: Addressing the Infrastructure Funding Shortfall of America's Airports*. n.d. URL: https://airportscouncil.org/intelligence/airport-infrastructure-needs-study/.

[12] R. Kuelpmann. *Addressing Climate Change Will Require Radical Changes: the 1.5-Degree Lifestyles Report is Out*. Mar. 2019. URL: https://www.cscp.org/1-5-degree-lifestyles-report/.

[13] Environmental Protection Agency. *Annual Concentration by Monitor (2025) [annual_conc_by_monitor_2020.zip]*. 2025. URL: https://aqs.epa.gov/aqsweb/airdata/download_files.html#Meta.

[14] N. Hudda et al. "Aviation Emissions Impact Ambient Ultrafine Particle Concentrations in the Greater Boston Area". In: *Environmental Science & Technology* 50.16 (2016), pp. 8514–8521. DOI: 10.1021/acs.est.6b01815.

[15] Jesse H Kroll et al. "The complex chemical effects of COVID-19 shutdowns on air quality". In: *Nature Chemistry* 12.9 (2020), pp. 777–779.

[16] E. Zender-Świercz et al. "The effect of temperature and humidity of air on the concentration of particulate matter—PM2.5 and PM10". In: *Atmospheric Research* 312 (2024), p. 107733. DOI: 10.1016/j.atmosres.2024.107733.